



AMC ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Capacity Planning for IT

Sub Code:BCS456B

MODULE-1

1. What was the primary focus of the book's first edition?
 - a. Capacity planning for cloud environments.
 - b. The transition to serverless architectures.
 - c. **Use of public clouds.**
 - d. Forecasting IT infrastructure spending.

Answer C

2. What is a significant benefit of serverless architectures mentioned in the text?
 - a. Reduced vendor dependencies.
 - b. **Reduced operational cost and complexity.**
 - c. Improved system performance at peak times.
 - d. Increased availability of digital media.

Answer B

3. Which metric is NOT mentioned as part of defining an application's load and capacity requirements?
 - a. Response times.
 - b. Consumable capacity.
 - c. Peak-driven processing.
 - d. **User satisfaction scores.**

Answer D

4. According to the text, what is an example of peak-driven processing?
 - a. Increased mobile traffic.
 - b. Daily routine operations of a web application.
 - c. **Spike in tweets during the Super Bowl.**
 - d. Continuous monitoring of system performance.

Answer C

5. What is one of the challenges of capacity planning in datacenters mentioned in the text?

- a. Short supply-chain cycles.
- b. High cost of elasticity.
- c. Long supply-chain cycles and configuration overhead.
- d. Ease of horizontal scaling.

Answer.C

6. Why is quick and dirty math important in capacity planning according to the text?
- a. It provides detailed and precise calculations.
 - b. It helps connect the dots between exact inputs and makes best guesses for the future.
 - c. It eliminates the need for horizontal scaling.
 - d. It ensures complete and unbiased information.

Answer b

7. What is a potential risk of lacking proper detail in capacity planning in the cloud?
- a. Over-provisioning leading to increased costs.
 - b. Under-provisioning leading to poor user experience.
 - c. Long supply-chain cycles.
 - d. Difficulty in setting up autoscaling policies.

Answer A

8. What should be included in the capacity-planning process as functionalities move from hardware to software?
- a. Increased server capacity.
 - b. Autoscaling policies.
 - c. Detailed user satisfaction surveys.
 - d. Software components.

Answer d

9. What are the three types of system failures mentioned in the text that are crucial for capacity planning?

- a) Software bugs, hardware malfunctions, network outages
- b) SLA violations, graceful degradation, true system failure
- c) High latency, data loss, security breaches
- d) User errors, system overloads, database corruptions

Answer: B. SLA violations, graceful degradation, true system failure

10. In the context of capacity planning, what does "graceful degradation" refer to?

- a) Complete system shutdown
- b) Slightly reduced performance under increased load
- c) Sudden failure without warning
- d) Manual intervention to prevent failure

Answer: B. Slightly reduced performance under increased load

11. Why is it important to tie server statistics to actual site or mobile app metrics?

- a) To reduce the need for hardware upgrades
- b) To better understand and track how capacity will change over time
- c) To ensure data privacy and security
- d) To simplify the architecture of the system

Answer: B. To better understand and track how capacity will change over time

12. Which factor is crucial when deciding where to focus priority attention in the capacity planning process?

- a) The number of servers in the system
- b) The cost of hardware
- c) The features that tax the capacity more than others
- d) The total data storage capacity

Answer: C. The features that tax the capacity more than others

13. How can tying system statistics to business metrics benefit non-technical stakeholders?

- a) It reduces the complexity of the system
- b) It makes it easier to secure funding for new hardware and software
- c) It helps in finding software bugs
- d) It ensures compliance with legal standards

Answer: B. It makes it easier to secure funding for new hardware and software

14.What is the primary goal of capacity planning according to the text?

- a) To reduce operational footprint and improve resource utilization
- b) To deliver the best end-user experience
- c) To ensure all systems run at maximum capacity
- d) To minimize costs associated with technology investments

Answer: B. To deliver the best end-user experience

15.Which approach is suggested to manage unexpected spikes in traffic without always running maximum capacity?

- a) Reducing the number of active servers
- b) Implementing queuing and caching-based approaches
- c) Decreasing the quality of service
- d) Limiting the number of users accessing the service

Answer: B. Implementing queuing and caching-based approaches

16.Why is collaboration with product and engineering teams crucial in capacity planning?

- a) To reduce the overall workload
- b) To align capacity requirements with long-term roadmaps and predict needs accurately
- c) To minimize the need for new hardware
- d) To ensure the security of the system

Answer: B. To align capacity requirements with long-term roadmaps and predict needs accurately

17.What is a potential risk of not having adequate capacity ready during traffic spikes?

- a) Increased operational costs
- b) Improved user experience
- c) Violations of SLAs and adverse impacts on user experience
- d) Reduced data redundancy

Answer: C. Violations of SLAs and adverse impacts on user experience

18. What is the purpose of using graphs and measurement tools in capacity planning?

- a) To simplify the architecture of the system
- b) To visually represent system performance and assist in decision-making
- c) To reduce the need for technical expertise
- d) To limit the amount of data collected

Answer: B. To visually represent system performance and assist in decision-making

19. In the context of cloud infrastructure, what does "cloud jail" refer to?

- A) A security issue where data is trapped and inaccessible.
- B) The scenario where a company is excessively spending on cloud services and is dependent on the cloud provider.
- C) A situation where cloud services are unavailable due to downtime.
- D) The restriction of cloud services to a single geographic location.

Answer: B) The scenario where a company is excessively spending on cloud services and is dependent on the cloud provider.

20. What advantage do smaller companies have in the procurement process compared to larger companies?

- A) They get better prices on equipment due to bulk buying.
- B) They can procure equipment faster due to a less cumbersome approval process.
- C) They have more negotiating power with vendors.
- D) They can access more advanced technology.

Answer: B) They can procure equipment faster due to a less cumbersome approval process.

21. What is the primary goal of performance tuning?

- A) To add new hardware and software to the system.
- B) To optimize the existing system for better performance.
- C) To predict future system requirements based on current usage.
- D) To benchmark the system in artificial environments.

Answer: B) To optimize the existing system for better performance.

22. According to the text, why should capacity planning be based on real-world observations?

- A) Because theoretical measurements are always inaccurate.
- B) Because empirical observations provide more relevant data for planning.
- C) Because benchmarking tools are outdated and unreliable.
- D) Because it is a requirement by cloud providers.

Answer: B) Because empirical observations provide more relevant data for planning.

23. What is a key challenge mentioned in the text for web operations with user-generated content?

- A) Predicting growth patterns due to user control over content.
- B) Ensuring data security and compliance.
- C) Maintaining website design aesthetics.
- D) Integrating with third-party services.

Answer: A) Predicting growth patterns due to user control over content.

24. In the example of the butcher in San Francisco, what is the preferred solution for increasing capacity?

- A) Tuning the existing trucks to go faster.
- B) Reducing the speed limit for trucks.
- C) Purchasing more trucks.
- D) Changing the route to San Jose.

.Answer: C) Purchasing more trucks.

25.What is one major impact of architecture on capacity management mentioned in the text?

- A) It determines the total cost of ownership for IT infrastructure.
- B) It affects how easily and flexibly you can add or subtract capacity.
- C) It dictates the physical location of data centers.
- D) It influences the color scheme and design of web applications.

Answer: B) It affects how easily and flexibly you can add or subtract capacity.

26.Why is it important to monitor API usage in web services?

- A) To ensure compliance with internal coding standards.
- B) To watch for emerging patterns and prevent system abuse.
- C) To track the popularity of the API among developers.
- D) To maintain the aesthetic quality of the website.

Answer: B) To watch for emerging patterns and prevent system abuse.

27.What is the relationship between performance tuning and capacity planning as described in the text?

- A) They are the same processes with different names.
- B) They have different goals but affect each other.
- C) Capacity planning always follows performance tuning.
- D) Performance tuning is unnecessary if capacity planning is done correctly.

Answer: B) They have different goals but affect each other.

28.What should be the focus when determining capacity needs according to the butcher example?

- A) Enhancing the speed of existing resources.
- B) Acquiring additional resources.
- C) Reducing the demand for resources.
- D) Changing the business model to fit existing resources.

Answer: B) Acquiring additional resources.